

Stuti Dewan

Software Engineer | ML & Data Systems | Python, C, Algorithms, Personalization

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EDUCATION

Indiana University Bloomington (IUB)

Bachelor of Science in Computer Science

Bloomington, IN

Aug 2021- May 2025

- Major with a specialization in Artificial Intelligence, Software Engineering, Data Science, and Minor in Data Science

TECHNICAL SKILLS

Languages: Java, Python, Swift (basic), JavaScript, C++, SQL

Test Development & Automation: PyTest, JUnit, Jest, unittest, API Testing, UI Testing, Integration Testing, Unit Testing, Regression Testing, Coverage Tracking

Tools & Frameworks: Jenkins, Git, Docker, Postman, VSCode, Unix Shell, Xcode (basic), Pygame

Development Practices: CI/CD Pipelines, Debugging, System Testing, Version Control, Code Quality Metrics

Core Strengths: OOP (Java, Python), Test Framework Design, Automation Scripting, REST APIs, OS Fundamentals, Networking, Tooling Infrastructure, Scalable Systems

Machine Learning: Logistic Regression, SVM, MLP, CountVectorizer, Cross-validation, Text Preprocessing, Feature Engineering, Model Evaluation

EXPERIENCE

NetSuite – Remote

May 2024 – Aug 2024

Software Training Intern

- Developed scalable Java automation modules to validate ERP workflows, reducing manual QA effort by 40% and improving regression stability across financial systems.
- Architected reusable test classes using OOP, enabling cross-team integration testing and faster onboarding for new features.
- Integrated automation scripts into Jenkins pipelines, streamlining CI/CD and enabling real-time deployment tracking.
- Collaborated with QA and development teams to triage flaky tests, contributing to a diagnostics dashboard and increasing sprint release confidence.

Management Science Associates - Pittsburgh, Pennsylvania

Software Development Intern

May 2023 - Aug 2023

- Engineered scalable backend data pipelines in Python/SQL, reducing processing latency by 75% and enabling near-real-time analytics across distributed systems.
- Built robust data validation frameworks to ensure schema integrity and quality assurance in high-throughput environments.
- Automated scenario testing for financial data using dynamic SQL triggers, boosting reporting accuracy and eliminating manual audits.
- Collaborated in cross-service debugging and optimization efforts, resolving integration bottlenecks and enhancing system fault tolerance.

Indiana University (IUB) - Bloomington, Indiana

Research Assistant

Aug 2022 - May 2023

- Audited the CSRankings algorithm for bias and transparency using Python-based test frameworks and reproducible ML evaluation pipelines.
- Engineered simulation tools to model ranking fluctuations under manipulated inputs, exposing vulnerabilities in conference-weighting heuristics.
- Designed and implemented a fairness test harness to benchmark algorithm robustness across parameter sets, contributing to more equitable ranking standards.

Marriott International - Singapore, Singapore

Software Developer Intern

May 2022 - Aug 2022

- Designed and deployed a statistical anomaly detection pipeline for transaction data, improving fraud detection accuracy by 15% using ML-based scoring logic.
- Built interactive debugging dashboards to visualize fraud scores and support compliance triaging across 50+ hotel properties.
- Developed modular Python test suites to verify transaction integrity, audit trail completeness, and model output consistency.
- Collaborated cross-functionally with engineering and compliance teams to productionize detection tools, maintaining a low false-positive rate.

ACADEMIC PROJECTS

Pathfinder: Python GUI App with Search Algorithms

- Built a Python app with a Pygame-based GUI to visualize 5+ pathfinding algorithms (A*, BFS, DFS, etc.) with random map generation and dynamic targets.
- Implemented debug overlays and runtime logs to evaluate algorithm efficiency under various map constraints.
- Concluded A with Euclidean heuristic* as the most balanced approach in terms of speed and accuracy compared to Greedy Search.

Sentiment Analysis on IMDB Movie Reviews Using Logistic Regression

- Built a sentiment classifier using logistic regression, achieving 85% accuracy on movie reviews with rigorous text preprocessing and lemmatization.
- Applied StratifiedKFold (5-fold) cross-validation and tracked performance consistency across folds.
- Developed a reusable text processing pipeline with vectorization using CountVectorizer, ensuring scalable feature extraction.

Breast Cancer Prediction Using Multi-layer Perceptron and Support Vector Machine Models

- Developed and compared MLP and SVM models on the UCI Breast Cancer dataset, optimizing MLP accuracy to 74%.
- Tuned hidden layer size and activation functions, analyzing tradeoffs between accuracy and computational cost.